Core Skills in Psychology and Neuroscience

Week 5 Schedule

03-11-2026

Hour 1

Introduction

- BaseR vs TidyR and the CRANiverse
- Guided tour around RStudio
  - console
  - environment
  - editor
  - files

Hello World

- The print function

Datatypes

- logical
- numeric
- text
- factor

Function calls

- Anatomy of a function call
- Input and ouput

Variables/objects

- <- vs = assignments
- scalers
- arrays
- matrices
  - tensors
- Indexing variables (square brackets)
- Lists (double square bracket)
- Dataframes
  - indexing dataframes

Break

Hour 2

Loops

- Curly brackets
- for loop
- while loops

Conditionals

- If statements
- Else statements
- ifelse()

Loading and saving files

- Delimiter files
- read.csv()
- write.csv()

Break

Hour 3

Installing and loading packages

- install.packages()
- library()
- require()

Reading and Writing Scripts

- Comments
- Scripted pipelines
  - load a file
  - manipulate it
  - save it

Break

Hour 4

Exploring statistics with R

- Data visualisation using BaseR

- Histograms
- Scatterplots
- summary()
- Random number generation